

Questions

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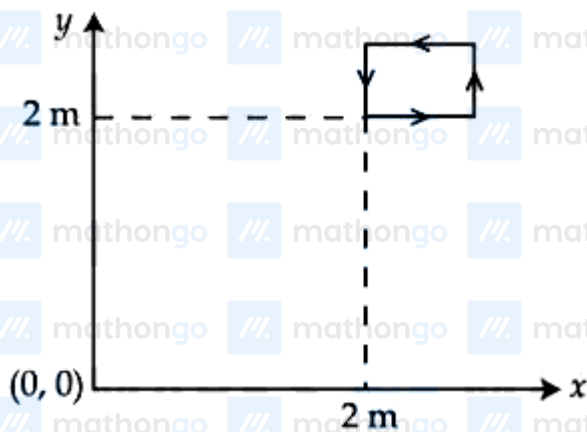
Q1 - 2024 (04 Apr Shift 1)

An electron is projected with uniform velocity along the axis inside a current carrying long solenoid. Then :

- (1) the electron will continue to move with uniform velocity along the axis of the solenoid.
- (2) the electron will be accelerated along the axis.
- (3) the electron path will be circular about the axis.
- (4) the electron will experience a force at 45° to the axis and execute a helical path.

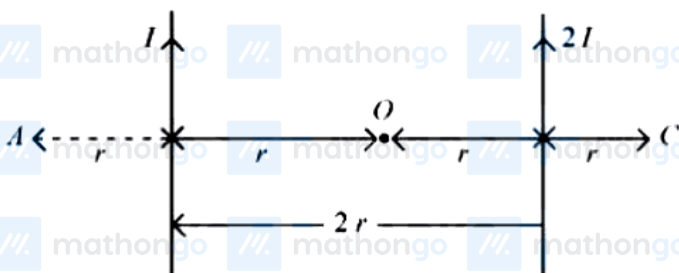
Q2 - 2024 (04 Apr Shift 1)

The magnetic field existing in a region is given by $\vec{B} = 0.2(1 + 2x)\hat{k}$ T. A square loop of edge 50 cm carrying 0.5 A current is placed in $x - y$ plane with its edges parallel to the $x - y$ axes, as shown in figure. The magnitude of the net magnetic force experienced by the loop is _____ mN.



Q3 - 2024 (04 Apr Shift 2)

Two parallel long current carrying wire separated by a distance $2r$ are shown in the figure. The ratio of magnetic field at A to the magnetic field produced at C is $\frac{x}{7}$. The value of x is _____



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Q4 - 2024 (05 Apr Shift 1)

In a co-axial straight cable, the central conductor and the outer conductor carry equal currents in opposite directions. The magnetic field is zero :

- (1) outside the cable
- (2) inside the outer conductor
- (3) inside the inner conductor
- (4) in between the two conductors

Q5 - 2024 (05 Apr Shift 1)

A 2 A current carrying straight metal wire of resistance 1Ω , resistivity $2 \times 10^{-6}\Omega\text{m}$, area of cross-section 10 mm^2 and mass 500 g is suspended horizontally in mid air by applying a uniform magnetic field $\vec{B}$. The magnitude of B is $\underline{\hspace{2cm}} \times 10^{-1}\text{ T}$ (given, $g = 10\text{ m/s}^2$).

Q6 - 2024 (05 Apr Shift 2)

The electrostatic force $(\vec{F}_1)$ and magnetic force $(\vec{F}_2)$ acting on a charge q moving with velocity v can be written :

- (1) $\vec{F}_1 = q\vec{E}, \vec{F}_2 = q(\vec{V} \times \vec{B})$
- (2) $\vec{F}_1 = q\vec{B}, \vec{F}_2 = q(\vec{B} \times \vec{V})$
- (3) $\vec{F}_1 = q\vec{E}, \vec{F}_2 = q(\vec{B} \times \vec{V})$
- (4) $\vec{F}_1 = q\vec{V} \cdot \vec{E}, \vec{F}_2 = q(\vec{B} \cdot \vec{V})$

Q7 - 2024 (05 Apr Shift 2)

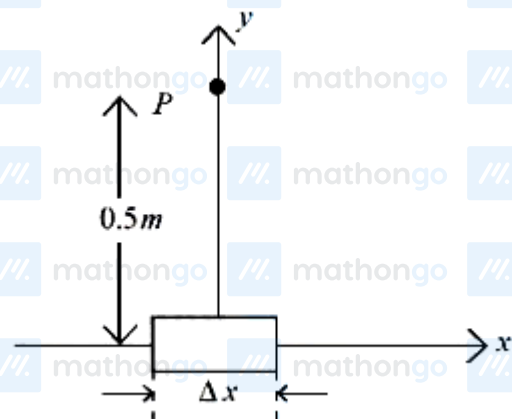
A solenoid of length 0.5 m has a radius of 1 cm and is made up of 'm' number of turns. It carries a current of 5 A. If the magnitude of the magnetic field inside the solenoid is $6.28 \times 10^{-3}\text{ T}$ then the value of m is $\underline{\hspace{2cm}}$.

Q8 - 2024 (06 Apr Shift 1)

Questions

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An element $\Delta l = \Delta x \hat{i}$ is placed at the origin and carries a large current $I = 10$ A. The magnetic field on the y -axis at a distance of 0.5 m from the elements Δx of 1 cm length is:



- (1) 4×10^{-8} T
- (2) 10×10^{-8} T
- (3) 8×10^{-8} T
- (4) 12×10^{-8} T

Q9 - 2024 (06 Apr Shift 1)

A circular coil having 200 turns, $2.5 \times 10^{-4} \text{ m}^2$ area and carrying $100 \mu\text{A}$ current is placed in a uniform magnetic field of 1 T . Initially the magnetic dipole moment ($\vec{M}$) was directed along $\vec{B}$. Amount of work, required to rotate the coil through 90° from its initial orientation such that $\vec{M}$ becomes perpendicular to $\vec{B}$, is _____ μJ .

Q10 - 2024 (06 Apr Shift 2)

A coil having 100 turns, area of $5 \times 10^{-3} \text{ m}^2$, carrying current of 1 mA is placed in uniform magnetic field of 0.20 T such a way that plane of coil is perpendicular to the magnetic field. The work done in turning the coil through 90° is _____ μJ .

Q11 - 2024 (08 Apr Shift 1)

Questions

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An electron with kinetic energy 5eV enters a region of uniform magnetic field of $3\text{ }\mu\text{T}$ perpendicular to its direction. An electric field E is applied perpendicular to the direction of velocity and magnetic field. The value of E , so that electron moves along the same path, is _____ NC^{-1} .

(Given, mass of electron $= 9 \times 10^{-31}\text{ kg}$, electric charge $= 1.6 \times 10^{-19}\text{ C}$)

Q12 - 2024 (08 Apr Shift 2)

A long straight wire of radius a carries a steady current I . The current is uniformly distributed across its cross section. The ratio of the magnetic field at $\frac{a}{2}$ and $2a$ from axis of the wire is :

(1) $1 : 4$

(2) $1 : 1$

(3) $3 : 4$

(4) $4 : 1$

Q13 - 2024 (09 Apr Shift 1)

Given below are two statements :

Statement (I) : When currents vary with time, Newton's third law is valid only if momentum carried by the electromagnetic field is taken into account.

Statement (II) : Ampere's circuital law does not depend on Biot-Savart's law.

In the light of the above statements, choose the correct answer from the options given below :

(1) Both Statement I and Statement II are true

(2) Statement I is true but Statement II is false

(3) Both Statement I and Statement II are false

(4) Statement I is false but Statement II is true

Q14 - 2024 (09 Apr Shift 1)

A square loop of edge length 2 m carrying current of 2 A is placed with its edges parallel to the x y axis. A

magnetic field is passing through the $x - y$ plane and expressed as $\vec{B} = B_0(1 + 4x)\hat{k}$, where $B_0 = 5\text{ T}$. The

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net magnetic force experienced by the loop is _____ N.

Q15 - 2024 (09 Apr Shift 2)

A proton and a deuteron ($q = +e, m = 2.0u$) having same kinetic energies enter a region of uniform magnetic field $\vec{B}$, moving perpendicular to $\vec{B}$. The ratio of the radius r_d of deuteron path to the radius r_p of the proton path is:

(1) $\sqrt{2} : 1$

(2) $1 : 1$

(3) $1 : \sqrt{2}$

(4) $1 : 2$

Q16 - 2024 (09 Apr Shift 2)

A straight magnetic strip has a magnetic moment of 44Am^2 . If the strip is bent in a semicircular shape, its magnetic moment will be _____ Am^2 .

(given $\pi = \frac{22}{7}$)

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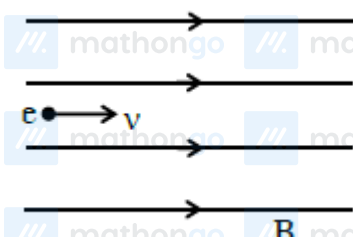
Answer Key

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Solutions

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Q1



Since $\vec{v} \parallel \vec{B}$ so force on electron due to magnetic field is zero. So it will move along axis with uniform velocity.

Q2

Force on segment parallel to x-axis will cancel each other. Hence F_{net} will be due to portion parallel to y-axis.

$$F = 0.5 \times 0.5 \times 6 \times 0.2 - 0.5 \times 0.5 \times 0.2 \times 5$$

$$= 0.5 \times 0.5 \times 0.2$$

$$= 0.25 \times 0.2$$

$$= 50 \times 10^{-3} \text{ N}$$

$$= 50 \text{ mN}$$

Q3

$$B_A = \frac{\mu_0 i}{2\pi r} + \frac{\mu_0 (2i)}{2\pi (3r)} = \frac{5\mu_0 i}{6\pi r}$$

$$B_C = \frac{\mu_0 (2i)}{2\pi r} + \frac{\mu_0 i}{2\pi (3r)} = \frac{7\mu_0 i}{6\pi r}$$

$$\therefore \frac{B_A}{B_C} = \frac{5}{7}$$

$$\therefore x = 5$$

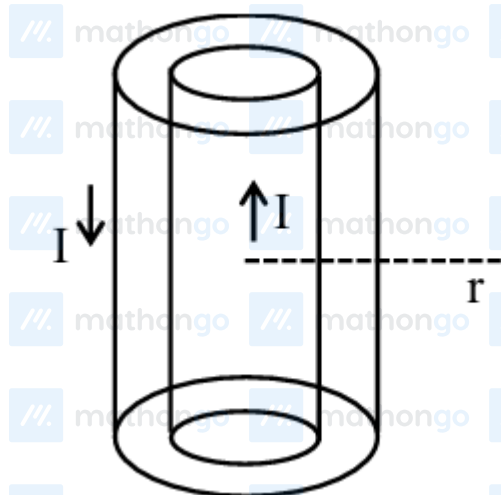
Q4

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Solutions

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$$\oint \vec{B} \cdot d\vec{\ell} = \mu_0 i_{\text{enc}} = 0$$

$\therefore B = 0$ outside the cable

Q5

$$R = \frac{\rho \ell}{A} \Rightarrow \frac{2 \times 10^{-6} \times \ell}{10^{-5}} = 1 \Rightarrow \ell = 5$$

$$mg = Bi\ell$$

$$B = \frac{mg}{i\ell} = \frac{5}{2 \times 5} = 0.5 = 5 \times 10^{-1} \text{ Tesla}$$

Q6

$$\vec{F}_1 = q\vec{E}$$

$$\vec{F}_2 = q(\vec{V} \times \vec{B})$$

(Theory)

Q7

$$\mu_0 n i = B \quad n = \text{number of turns per unit length}$$

$$\mu_0 \left(\frac{m}{\ell} \right) i = B$$

$$m = \frac{B \cdot \ell}{\mu_0 i} = \frac{6.28 \times 10^{-3} \times 0.5}{12.56 \times 10^{-7} \times 5}$$

$$m = 500$$

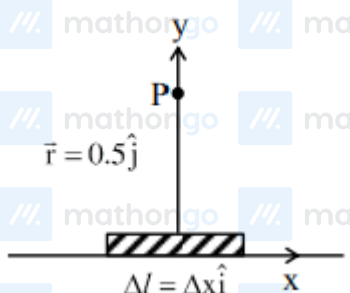
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Solutions

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$$\vec{dB} = \frac{\mu_0 I}{4\pi} \frac{(\vec{dl} \times \vec{r})}{r^3} \text{ (Tesla)}$$

$$= \frac{10^{-7} \times 10 \times \left(\frac{1}{2} \times \frac{1}{100}\right) (+\hat{k})}{\left(\frac{1}{2}\right)^3} = 4 \times 10^{-8} \text{ T} (+\hat{k})$$

Q9



We know

$$\begin{aligned} W_{\text{ext}} &= \Delta U + \Delta KE \quad (\text{P.E.} = -\vec{M} \cdot \vec{B}) \\ &= -\vec{M} \cdot \vec{B}_f + \vec{M} \cdot \vec{B}_i + 0 \\ &= -MB \cos 90^\circ + MB \cos 0^\circ \\ &= MB \\ &= NIAB \\ &= 200 \times 100 \times 10^{-6} \times \frac{5}{2} \times 10^{-4} \times 1 = 5 \mu\text{J} \end{aligned}$$

Q10

$$\begin{aligned} W &= \Delta U = U_f - U_i \\ W &= (-\vec{\mu} \cdot \vec{B})_f - (-\vec{\mu} \cdot \vec{B})_i \\ &= 0 + (\vec{\mu} \cdot \vec{B})_i \\ &= (100 \times 5 \times 10^{-3} \times 1 \times 10^{-3}) \times 0.2 \text{ J} \\ &= 1 \times 10^{-4} \text{ J} = 100 \mu\text{J} \end{aligned}$$

Q11

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Solutions

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For the given condition of moving undeflected, net force should be zero.

$$qE = qVB$$

$$E = VB$$

$$= \sqrt{\frac{2 \times KE}{m}} \times B$$

$$= \sqrt{\frac{2 \times 5 \times 1.6 \times 10^{-19}}{9 \times 10^{-31}}} \times 3 \times 10^{-6}$$

$$= 4 \text{ N/C}$$

Q12

$$B_1 2\pi \frac{a}{2} = \mu_0 \frac{I}{4}$$

$$B_1 = \frac{\mu_0 I}{4\pi a}$$

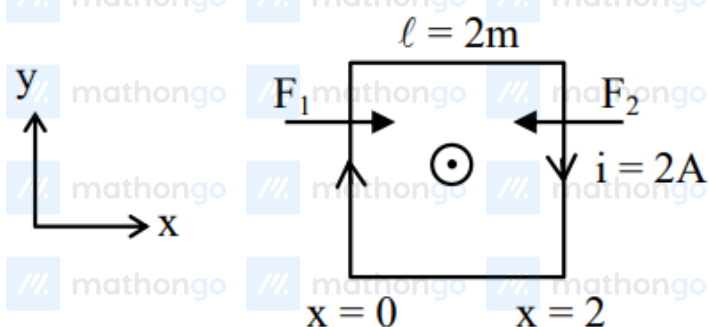
$$B_2 2\pi 2a = \mu_0 I$$

$$B_2 = \frac{\mu_0 I}{4\pi a}$$

Q13

Conceptual.

Q14



$$B(x = 0) = B_0,$$

$$B(x = 2) = 9 B_0$$

$$\text{Also, } F = i\ell B$$

$$\Rightarrow F_1 = i\ell B_0 \text{ \& } F_2 = 9i\ell B_0$$

$$F = F_2 - F_1 = 8i\ell B_0 = 8 \times 2 \times 2 \times 5$$

$$F = 160 \text{ N}$$

Q15

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Solutions

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In uniform magnetic field,

$$R = \frac{mv}{qB} = \frac{\sqrt{2m(K.E)}}{qB}$$

Since same K.E

$$R \propto \frac{\sqrt{m}}{q}$$

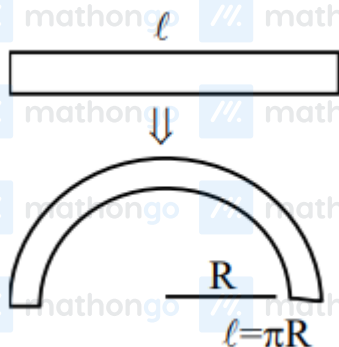
$$\therefore \frac{R_{\text{deuteron}}}{R_{\text{proton}}} = \sqrt{\frac{m_d}{m_p}} \times \frac{q_p}{q_d}$$

$$= \sqrt{2} \times 1$$

$$\therefore \gamma_d : \gamma_p = \sqrt{2} : 1$$

Q16

Magnetic moment of straight wire = $mx\ell = 44$



Magnetic moment of arc

$$= m \times 2r$$

$$= m \times \frac{2\ell}{\pi}$$

$$= \frac{44 \times 2}{\pi} = \frac{88}{\pi} = 28$$

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